

Real-World Case Study

Manchester Arena Bombing, 2017

Good intentions. Zero coordination. Wasted donations.



Thousands queued

Citizens rushed to donate blood in a wave of public goodwill

Hospitals turned donors away

Bloodbanks were overwhelmed — many has more than enough

No system to coordinate

Without a triage layer, good intentions became a logistical crisis

The problem wasn't unwillingness to help — it was the absence of a coordination layer



BloodLine

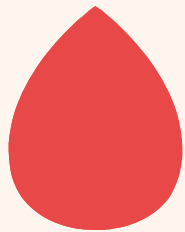
The right blood, at the right place, **at the right time**

Team Name: 404

Team Members: Sia Hui Qi, Sia Wen Xin, Tan Lu Xing Neston

Our Solution – BloodLine

A digital traffic controller for emergency blood donation.



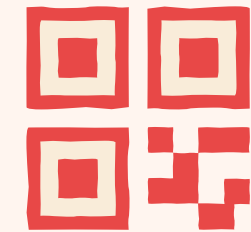
Live Inventory

Blood banks push
real-time stock levels
per type to a central
system



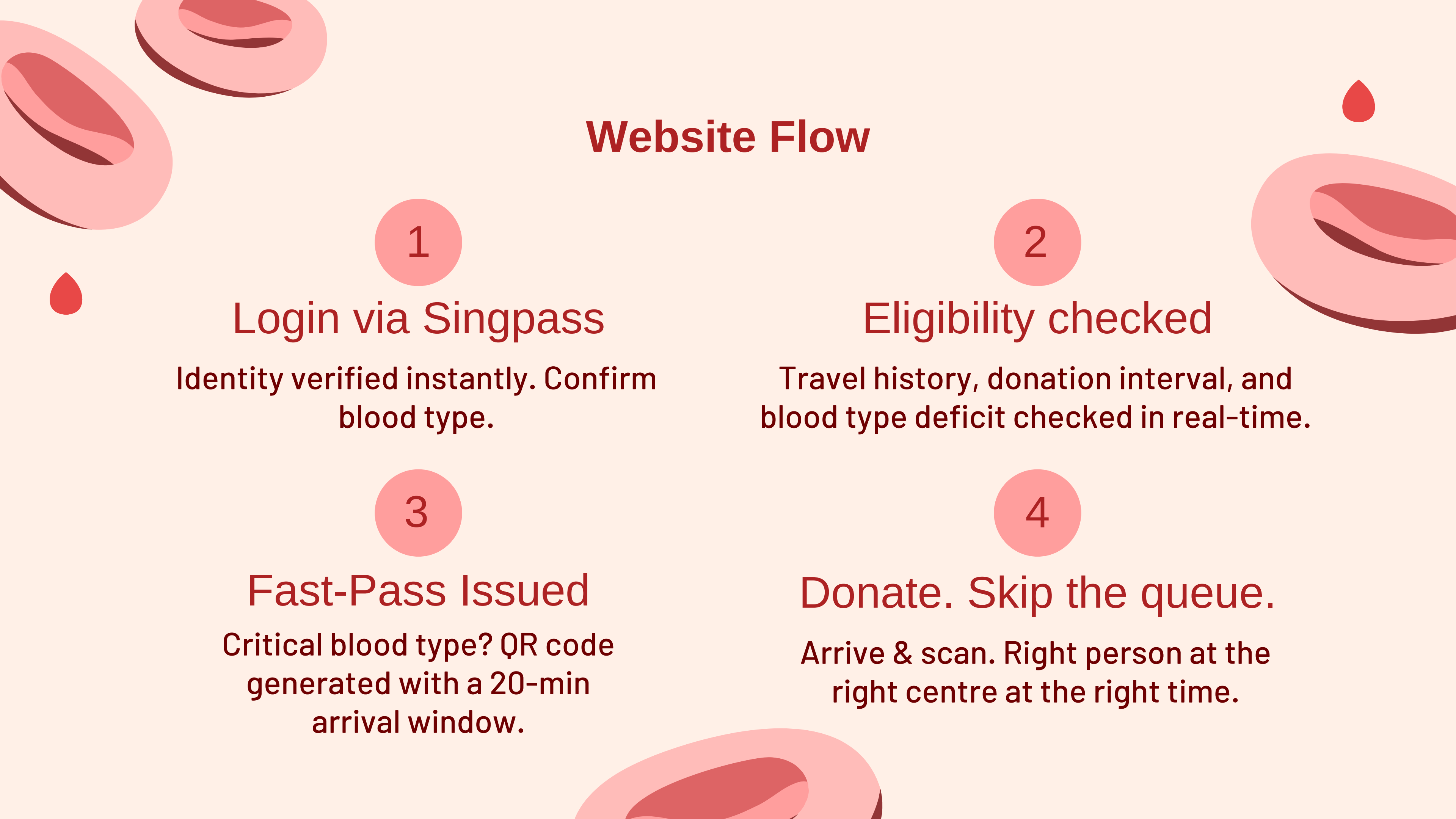
Singpass Identity

Donors login via
Singpass — blood
type & eligibility
verified instantly



Fast-Pass QR

Critical donors get a
QR code with a strict
20-min arrival window



Website Flow

1

Login via Singpass

Identity verified instantly. Confirm blood type.

2

Eligibility checked

Travel history, donation interval, and blood type deficit checked in real-time.

3

Fast-Pass Issued

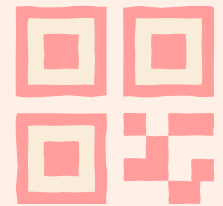
Critical blood type? QR code generated with a 20-min arrival window.

4

Donate. Skip the queue.

Arrive & scan. Right person at the right centre at the right time.

Key Features



Dynamic Fast-Pass

QR code with strict 20-min window. Bypasses all queues



Live blood inventory

All centres update stock continuously.



Singpass verified

Identity pulled directly. Zero self-reported risk.



Travel deferral engine

HSA-aligned rules auto-remove ineligible donors.



Rewards system

Points, tiers, vouchers



Why BloodLine works.

WITHOUT BLOODLINE

✗ 4-hour queues at blood banks.

✗ Critical types run out mid-crisis.

✗ Eligible donors don't know where to go.

✗ Ineligible donors clog the system.



WITH BLOODLINE

✓ 20-min Fast-Pass windows – no crowds.

✓ Sort donors by exact blood type needed.

✓ Donors directed to blood bank of choice.

✓ Travel deferral engine removes ineligible silently.





**When every minute counts,
BloodLine delivers.**

20 min

Fast-Pass arrival
window

8

Blood types
supported

2

Portals – donor +
admin

